

Model Conversion and Quantization Analysis Report

Model Size Compression Analysis

A significant reduction in model footprint is the primary prerequisite for deploying neural networks on edge devices such as microcontrollers. Based on the physical file sizes generated during the conversion process, the storage comparison is as follows:

Model Format	File Size	Compression Ratio	Size Reduction
Original Keras (.keras)	286.33 KB	Baseline	/
Standard TFLite (Float32)	88.86 KB	3.22x	68.97%
Quantized TFLite (Int8)	25.94 KB	11.04x	90.94%

Converting the original Keras model to a standard Float32 TFLite format immediately reduced the size by nearly 69%. This is because the TFLite converter optimizes the computation graph by stripping away training-specific, redundant nodes (such as optimizer states and backpropagation gradient graphs), leaving a lightweight structure strictly optimized for inference.

By introducing Full Integer (Int8) Quantization, the model size was further compressed to 25.94 KB, achieving an overall reduction of 90.94%. This leap in efficiency is driven by the data type down casting: all weight parameters and activation values were mapped from 32-bit floating-point numbers (4 bytes) to 8-bit integers (1 byte).

Model Accuracy and Performance Impact

Quantization is a form of "lossy compression" and typically introduces a slight drop in model precision. However, the evaluation on the MNIST test set demonstrated excellent robustness:

Model Format	Test Set Accuracy	Accuracy Delta
Standard TFLite (Float32)	97.80%	Baseline
Quantized TFLite (Int8)	97.81%	+0.01%

Interestingly, the quantized model did not show a drop in accuracy during the test. Instead, it saw a tiny improvement of 0.01%. This is actually a normal outcome. When we convert detailed floating-point numbers into simpler integers, the process can naturally filter out some minor noise in the data, which sometimes helps the model perform slightly better on the test set. Overall, this result simply confirms that an 8-bit format is perfectly sufficient to handle image classification tasks like MNIST without losing important features.

Conclusion

By applying TensorFlow Lite conversion and Int8 Full Integer Quantization, the system successfully reduced the neural network's storage footprint by nearly 91% (down to 25.94 KB) while maintaining an excellent predictive accuracy of 97.81%. This conversion clearly demonstrates the massive spatial advantages of quantization techniques, proving that the model can be efficiently optimized for resource-constrained environments without sacrificing performance.